

WASP-43b

R-2058

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_151221 · created 2026-08-09T00:54:26+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457377.8936 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.057
Transit depth (Rp/Rs)^2	3.87 +/- 2.23 [%]
Orbital Inclination (inc)	81.53 +/- 2.81
Airmass coefficient 1 (a1)	1.39 +/- 0.48
Airmass coefficient 2 (a2)	-0.18 +/- 0.14
Scatter in the residuals of the lightcurve fit is	33.3 %
Best Comparison Star	None
Optimal Aperture	3.01
Optimal Annulus	7.51
Transit Duration (day)	0.048 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.197 ± 0.057 vs 0.1594 (deviation 0.64σ)
Duration fit vs published	0.048 d vs 0.0512 d (deviation 0.22σ)
Mid-time O–C	-2.4 min
Depth SNR	3.3
Residual scatter	33.3 %

Light curve

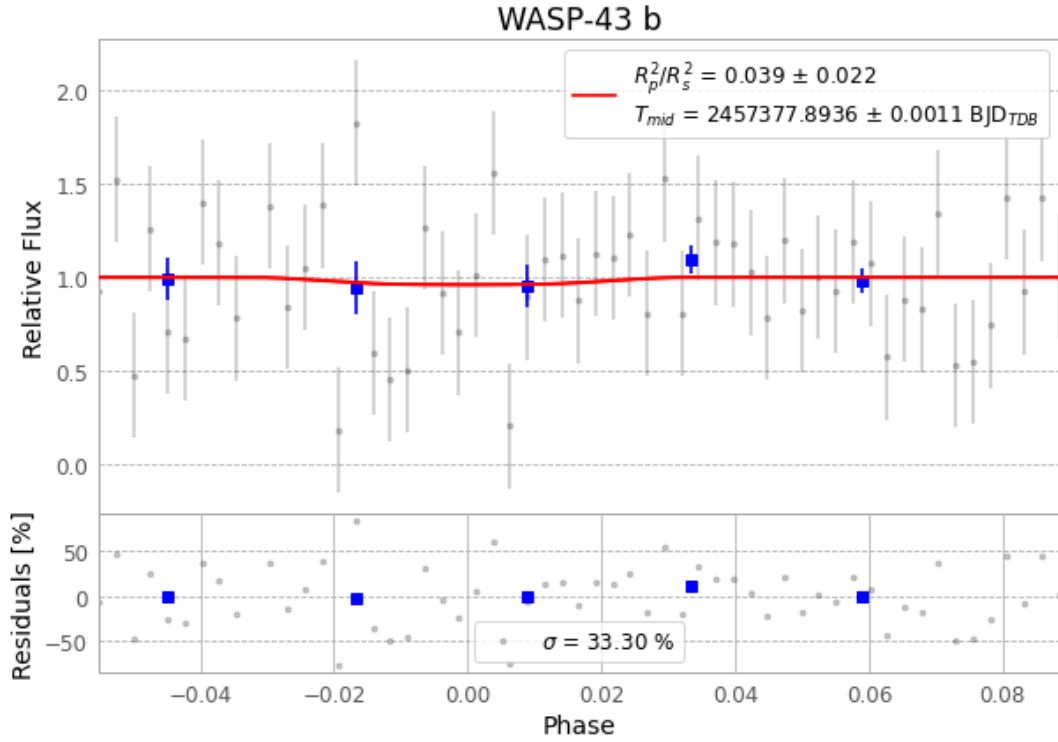


Figure 1 — FinalLightCurve_WASP-43 b_2015-12-21.png (SHA-256 a572ea420c2adf02...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [316, 154], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7d2decc9ad345464...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

57 FITS frames (37 MB), WASP-43151221081812.FITS → WASP-43151221110612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-151221.log (545b32dc0dc5...), FinalLightCurve_WASP-43 b_2015-12-21.png (a572ea420c2a...), FinalLightCurve_WASP-43 b_2015-12-21.csv (12a78a08b5fa...)

Compute resources

Wall time 237.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 00:54Z.